Julià Delos Ayllón

Senior Electrical Engineer · Entrepreneur

03/03/1983 - Mataró, Spain

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Profile

Spanish seasoned engineer and failed entrepreneur with a PhD in Microelectronics, bringing together a deep understanding of hardware and software and a passion for building things. Enjoys turning theory into working systems and uses Blender to bring personal engineering and 3D printing projects to life. Eager to contribute this experience, curiosity, and Mediterranean passion to Blender's development.

Experience

Founder — *Netherlandis*
La Calçotada — *2019 – 2026*

- Turned a yearly food festival into a food-experience business serving up to **10,000 users** across the Netherlands and seven neighbouring countries, building the e-commerce flow, logistics and supplier network.
- Generated **EUR 100k+** in online sales during the initial market validation phase, reaching a peak run-rate of around **EUR 20k/month**, with gross margins above **40%** and approximately **3%** e-commerce conversion.
- Partnered with restaurants in Amsterdam, Eindhoven and Tilburg to host around **200 guests** per afternoon during otherwise quiet or closed hours, generating approximately **EUR 19 per guest** for each restaurant.
- Built an equity-crowdfunding platform intended to give the community ownership of the business, applying open-source principles to an open business model; the initiative failed at the go-to-market stage.

Co-Founder and CTO — *Remote (team in Portugal)*
Ripatrip — *Sept 2019 – June 2020*

- Co-founded a travel search platform supported by **Startup Lisboa**, initially focused on kitesurfing and later expanded to ski and snowboard destinations.
- Delivered a working demo within **two weeks** and presented it at **Web Summit 2019 in Lisbon**. Guided **two junior developers** through further development.
- Enabled travel searches across **5,000 destinations** by integrating multiple APIs for destination data and travel offerings.
- Desinged a servrless architecture with the statice site generator Gatsby and Redux, and a headless CMS; using Netlify hosting infrasturcutre

Innovation and Technology Consultant — *Eindhoven, Netherlands*
ICT Group — *Sept 2016 – Sept 2018*

- Worked closely with the C-suite to introduce visual business modelling tools across business units, creating a shared language for strategy and innovation.
- Convinced senior leadership to establish a **dedicated data-science team and data lake** by showing business-unit leaders how data science could benefit their units.
- Coached around **10 developer-led teams** for the internal Dragons' Den, using experiments to validate technical ideas and customer value.

Research Scientist — *Eindhoven, Netherlands*
Philips Research — *June 2011 – Dec 2015*

- Surpassed state-of-the-art integrated power density by developing a new power-system-on-chip LED driver.
- Enabled chip-area optimization by creating a design framework based on a closed-form model, validated theoretically and experimentally.
- Expanded Philips' patent portfolio through circuit inventions resulting in **8 first-named EPO patent records (6 granted)**.
- Completed a PhD with Eindhoven University of Technology, publishing peer-reviewed research and presenting internationally.

Hardware Engineer — *Valls, Spain*
Lear Corporation — *Mar 2010 – May 2011*

- Led a **60 kW** racing-car inverter project, coordinating software and mechanical engineers, university partners, and a technical consortium. Delivered on time, meeting size and performance targets initially considered infeasible.
- Led hardware design, testing, and production, delivering an inverter that worked on its first iteration.
- Set up a high-voltage, high-power test lab for testing above 100 kW and up to 2 kV, defining equipment requirements and operating procedures for safe testing.
- Reduced current-sensor cost to **one-sixth** by designing an in-house, low-profile tangential sensor with adjustable sensitivity and high accuracy.

Embedded Systems Engineer, Printer Firmware *Barcelona*
HP (via Capgemini) *Sept 2006 – Sept 2007*

- Delivered production fixes within the **first month** in a large C/C++ and Java codebase targeting a main Linux/OS and multipe distributed ASICs
- Resolved longstanding firmware issues by applying hardware expertise beyond the software team's capabilities.
- Managed builds, release packaging, and security updates for HP's top-of-the-line large-format printers using ClearCase.
- Transferred firmware knowledge to the Bangalore team through **two-to-three weeks** of hands-on training.

Education

Eindhoven University of Technology *2016*
PhD in Electrical Engineering

Technical University of Catalonia *2010*
MSc in Electronics Engineering

Polytech Lille *2006*
BSc in Industrial Software Development

Technical University of Catalonia *2005*
BSc in Electrical Engineering

Independent Projects

Switched-capacitor converter impedance tools *June 2026 – present*

- Web application of the power converter simulator of my PhD research, I am porting MATLAB to C++ (SymEngine) targeting WASAM.

Onion Memorandum platform *July 2024 – Mar 2025*

- Remix (Epic Stack) application with Prisma migrations, MDX content and PDF generation, tested with Vitest and Playwright and deployed with Docker on Fly.io via GitHub Actions.

QR ticketing app and ticket reader *Nov 2023 – Mar 2024*

- Shopify app that issues QR-code tickets as PDFs, plus a Remix scanner app built on XState; tested with Cypress and Vitest, deployed with Docker on Fly.io.

Renault Technocentre, LAGIS *France*
Embedded and real-time internships *Feb 2005 – July 2006*

- Benchmarked vehicle-speed estimators on dSPACE real-time hardware with MATLAB/Simulink (Renault).
- Wrote a CAN network sniffer in C on Linux/RTAI for the Robucar vehicle and the developer manual for SynDEx, a tool for distributed real-time embedded systems (LAGIS).

Hacklab Nòmada, Hackmeetings, Mataró Sense Fils *Mataró, Spain*
Hacktivism and community computing *1999 – 2009*

- Rebuilt discarded computers for people without access to technology and maintained a public computer room with free internet access.
- Learned Linux hands-on by setting up a server and network-booted computers, and helping run Linux community events.
- Co-organized a Hackmeeting in Mataró and volunteered at another in the Basque Country, each hosting around 400 people, supporting electrical setup and catering.

Skills

Programming: C, C++, Type/JavaScript, Python, Java, Solidity, MATLAB

Systems: Embedded Linux, firmware, ARM, real-time systems (dSPACE, RTAI), CAN, edge computing, WebAssembly

Web development: HTML, CSS, Ja React, Next.js, Remix, Astro, Gatsby, Django, Redux, XState, Prisma, Sanity, Shopify Remix Apps

DevOps: Linux administration, Docker, CI/CD with GitHub Actions, deployment workflows, self-hosted services, networking, reverse proxies, Git, ClearCase, Fly.io

Testing: Vitest, Playwright, Cypress, hardware/software integration testing

AI-assisted development: Pi agent, Hermes, Codex, Claude-Code, llama.cpp, Ollama, Mastra, human-in-the-loop agent workflows

Blockchain: Hardhat, OpenZeppelin, Viem, Ethers, smart-contract testing and DApp integration

Creative: Figma, Procreate, Blender, FreeCAD, Davinci Resolve Studio , Sharp3D

Languages: Native Catalan and Spanish · Proficient in English and French · Conversational in Portuguese and Dutch

Extracurricualr Activities: Tinkering with my 3D printer, enjoying the outdoors through climbing, kitesurfing and ski touring, and exploring the possibilities of slow cooking for family and friends.

Certifications and Grants

Strategyzer Academy: Business Models that Work and Value Propositions that Sell, 2026

La Caixa Master's scholarship: Full tuition and monthly stipend, 2007

Patent portfolio

EP2989714B1 · 2016: Compact power conversion device with continuous output regulation range
EP3011805B1 · 2018: Light emitting diode driver with differential voltage supply
EP3047564B1 · 2019: Compact driver, notably for a light emitting diode, having an auxiliary output
EP3047565B1 · 2019: Compact driver, notably for a light emitting diode, having an integrated dual output
EP3243264B1 · 2021: Power conversion device
EP3406109B1 · 2020: A driver and method for driving at least two sets of solid state lighting elements
EP3111726A1 · 2017: Driver circuit with extended operation range
EP3243265A1 · 2017: Power conversion device